6, 7, 14, 27

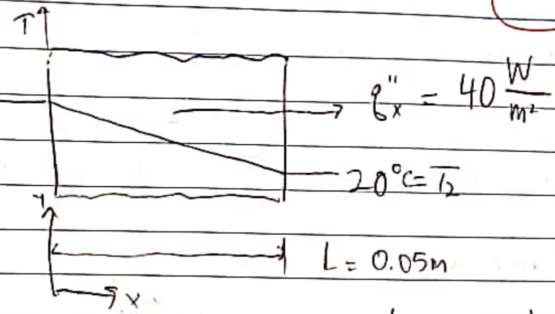
Japheth Bandi

80

CHE 312 HW 2

1/27/20

1.6



steady state

Fourier's law 1.2 $\therefore k = q''_x \frac{L}{T_1 - T_2} = 40 \frac{W}{m^2} \frac{0.05m}{(40-20)°C}$

$k = 0.10 \frac{W}{m \cdot K}$

1.7 1. heat flowing perpendicular to cross-sec-area

2. temps on both walls uniform.

conduction heat law $Q = \frac{KA \Delta T}{L}$

electrical analogy $Q = \frac{\Delta T}{R_{th}}$, $R_{th} = \frac{L}{KA}$

↳ thermal resistance.

find heat loss.

$= 1.19 \times 10^{-3} \frac{K}{W}$

$Q = \frac{10}{1.19 \times 10^{-3}}$ $\therefore Q = 8403.36\ W$

8,403 kW

$\therefore R = \frac{L}{KA}$ $k = \frac{0.005}{1.19 \cdot 10^{-3} \cdot 1.3} = 1.4 \frac{W}{m \cdot K}$

$R_{wcond} = \frac{L}{...}$ $\Rightarrow \frac{(0.005m)}{...} = (1.19 \cdot 10$

14, 27

1.14) convective heat flux $\left(\frac{\text{energy}}{m^2}\right) \rightarrow h(\Delta T)$ 0)

interior to wall, flux 1 $= 5(20-16) = 20$ W-m^{-2} Steady state.

wall "exterior" 2 $= 20(6-5) = 20$ W/m^2

heat entering = heat leaving wall bc flux1 = flux2

Missing? one calc

1.27.) a) $T_s = 300K$ $A = \frac{\pi D^2}{4} = \frac{\pi (0.3m)^2}{4} = 0.07068m^2$

$T_{sur} = 77K$

$\varepsilon = 0.25$

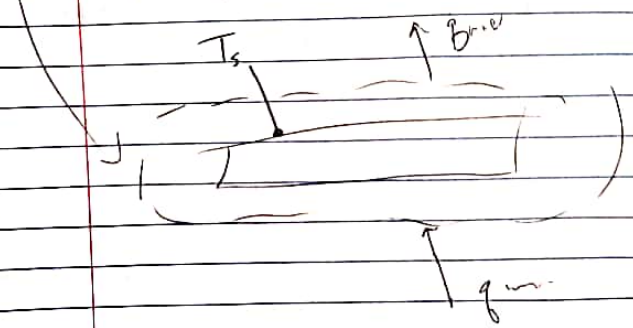
$\sigma = 5.67 \times 10^{-8} \frac{W}{m^2 K^4}$

$q = \varepsilon \sigma A (T_s^4 - T_{sur}^4) = \frac{1}{4}(5.67 \times 10^{-8} \frac{W}{m^3}) \cdot K^4 \cdot 0.07068 m^2 \times (300^4 - 77^4) K^4$

$q = 8.08 W$

b) Rate of N req'd $= 8.08 \cdot 3600 / 125 \cdot 10^3 \frac{kg}{hr} = 0.23 \frac{kg}{hr}$

c) tot. heat loss can be minimized bc reduction is in emissive. putting an aluminium sheet reduces liq. N cons.



Scanned with CamScanner